WILLAMETTE UNIVERSITY | MBA

SYLLABUS · FALL 2026

GSMPR-622A-S: Data, Statistics & Decision Science

Robert W. Walker · Fall 2026 · Salem cohort · Version of October 10, 2026

Course information

GSMPR-622A-S (Salem) · Willamette University · Atkinson Graduate School of Management · MBA for Professionals · Fall 2026

Meeting	When	Where
In person	**Tuesdays, 6:00–9:50 pm**	**Kaneko 120, Salem campus** · Google Meet: meet.google.com/dvx-rkng-stn
Online, with the Portland cohort (622A-P)	Thursdays, 6:00–8:15 pm (may run to 8:30)	Google Meet: meet.google.com/knh-gxyw-uwr

Instructor: Robert W. Walker · rwalker@willamette.edu

Office hours: my calendar is open. Book a Google Meet any time, or catch me before and after class. I will reach out to you first.

Course site: robertwwalker.github.io/Integrated622 for the schedule, session plans, readings and reading guides, and the shared dataset; Canvas for assignments and submissions.

Materials: no textbook to buy. Start with the Getting Started guide on the Canvas Start Here page. Readings and their reading guides are on the course site, one page per module; reserve readings are on Canvas under Files by Module. Expect about two hours of reading before each session, and keep your guide answers in one running document, your reading notebook. You will need Gemini, signed in with your Willamette account (see Policies).

Course description

Data is the measurement of models. We do not collect data and then wonder what it means. We build models of how organizations work, and data is how we find out whether those models are telling us the truth. There's also forecasting: models provide us with exactly what we need to forecast our uncertain world in a careful and coherent way.

An MBA is strategy built from models (finance, accounting, operations, organizations, marketing, HR, information systems, and data), and because a statistical model is itself a model of how the data came to be, every project here holds two models: one of the organization and one of its data.

Models do two jobs: they let us test what we believe against data, and they let us forecast what will happen if we act. Managers live on the second, and simulation and AI now put it within reach.

This course is an integration. You arrive from organizational management or operations management with a partially formed project: a model of how some part of an organization works. This course gives you the means to test it. You will define what to measure, find or honestly simulate the data, analyze it, and reason carefully about what the evidence can and cannot establish.

Three modules build one project:

1. **Data in Organizations** — what data is, where it lives in public, private, and nonprofit organizations, and how to turn a concept into a measurable variable.
2. **Analytical Techniques** — visualization, grouping, and association: what each can answer, what it cannot, and the difference between a model that explains and one that predicts.
3. **Causation** — Judea Pearl's framework for the question every manager needs answered: if you change something, how would you know it was your change that caused what followed?

No formulas, and no code. AI tools do the technical execution. Your work is the thinking: what to ask, what the answer means, and what the data cannot tell you. That is harder than the arithmetic, and it is work your prior course has prepared you for.

This course is also its own case. It is a redesign with a smaller classroom footprint, two simultaneous changes, and an imperfect baseline. We face exactly the question we will teach you to ask: did it work? That uncertainty is mine to carry, not yours.

Learning objectives

By the end of the course, you will be able to:

1. **State a model** of how part of an organization works, with its key concepts and the relationships you believe hold between them.
2. **Operationalize** a concept: translate it into a measurable variable, name what kind of variable it is, and say what it misses.
3. **Locate data** in an organization's systems, or specify an honest simulation of what the data would look like if your model is right.
4. **Recognize the ways data goes wrong** (missing, mismeasured, redefined, selected, mistimed) and say what to do about them.
5. **Describe and analyze** data with visualization, grouping, and association, and say what each finding does and does not establish.
6. **Reason causally** with Pearl's ladder and causal diagrams: name the confounders, and judge which rung your evidence reaches.
7. **Forecast** with a model: run it forward under scenarios to compare decisions, and say how uncertain the forecast is.
8. **Work with AI as an analyst**: direct it in plain language, check its work, and catch it when it is wrong.
9. **Integrate** your prior discipline's frameworks with data work in one coherent project.

How the course works

Each module follows the same rhythm: an in-person kick-off on Tuesday (6:00–9:50 pm), a Thursday online session, a second in-person Tuesday, and a second Thursday online session. Module 3 is split by Thanksgiving.

- **In-person nights** are where the ideas are taught and where you work on your project with your pod. Expect short activities on paper before anything on a screen.

- **Thursday online sessions** are hands-on: working with AI on a shared dataset and on your own project. Both cohorts meet together. Every Thursday session is recorded, and every Thursday activity has a self-paced version. If you cannot attend live, you can still do all of it. Pod posts are due by Sunday night, with one reply to a pod-mate. On October 29 we don't meet live: the Atkinson Community Roundtable hosts Oregon's Chief AI Officer that evening, and that session is recorded and posted by 8:00 pm.
- **Deadlines.** Project stages are due on in-person nights; module exercises are due on Sundays. Submit all work on Canvas under Assignments: work due on an in-person night by 6:00 pm that day, and Sunday work by 11:59 pm Pacific. Two exceptions at the end: the pre-final version of your white paper and slide deck is due Thursday, December 3, by 6:00 pm, and the final project is due Friday, December 4, by 11:59 pm Pacific.

Pods. You will be placed in a pod of about four classmates from your own location and program track, chosen by the teaching team for project similarity. Your pod reads your work first, gives feedback, and hears your dry run. Pods support; they do not co-author. Every project is individual.

AI as your analyst. We use AI throughout, in plain language, with Gemini as the course standard. AI produces the charts, statistics, and simulated datasets. You decide what question matters, what the result means, and what it cannot tell you. Learning to direct and check an AI analyst is a stated objective of this course.

The project

One individual project, built across all three modules and due Friday, December 4. It takes a model of some part of an organization, applies your discipline's lens to it, and brings real data or honest simulation to bear on what the model predicts.

The organization can be your workplace, a representative organization in a sector you know, or a useful fiction. **The data** can be real or simulated. Simulation is not fabrication: it is a rigorous tradition in supply chain, risk, and public health, and the intellectual work is in designing the model. Both paths are equally legitimate.

The project is built like a research paper, one section per module:

Stage	Section it becomes	What it contains	Due
Project Brief	Theory	Six elements, as described in Getting Started: organization, model, question, prediction, data sketch, disciplinary anchor (1–2 pages)	Tue Oct 20, 6:00 pm
Data & Methods draft	Data & Methods	Unit of observation, variable table, data source, quality concerns, first description of key variables (1–2 pages)	Tue Nov 3
Results draft	Results	At least one visualization, an association finding, and what it does and does not establish	Tue Nov 17
Dry run	—	A run-through of your presentation with your pod: model, data, claim, limits	Tue Dec 1
Pre-final version	—	Your white paper and slide deck, nearly finished, shared with your pod with comment access and submitted on Canvas, for review in the final session	Thu Dec 3, 6:00 pm
Final project	Interpretation, plus the whole	A recorded presentation (about 15 minutes), its slide deck, and a 2–4 page white paper: your causal claim, the rung of Pearl's ladder your evidence reaches, what the next step would require, and a forecast for the decision you recommend	Fri Dec 4, 11:59 pm

Each draft becomes a section of your white paper, so it is not extra work. You will revise as you go.

The final deliverable has three parts, all due Friday, December 4, by 11:59 pm Pacific:

1. **A video presentation of about 15 minutes.** Record yourself presenting your slides; Google Meet or Google Slides recording works. Upload the video to Google Drive with your Willamette account and submit the share link on Canvas. Walk through the model, the data, what you found, and what it does and does not establish.
2. **The slide deck** you presented.
3. **A white paper of 2–4 pages**, written for a decision-maker in your organization: the question, the model, the evidence, your causal claim and its limits, and the decision you recommend, with a forecast of what will happen if it is adopted: a range, and the assumptions behind it.

The pre-final version is due Thursday, December 3, by 6:00 pm: your white paper and slide deck, nearly finished. Share both with your pod with comment access and submit both links on Canvas; your pod reviews them in that night's session, and you revise before the final. The video is not part of the pre-final version. It is worth 5 points: 4 for being complete and on time (both the white paper and the slide deck: 4; only one: 2; neither: 0) and 1 for its content, judged against three of the final project's criteria: your causal claim and its rung, your recommendation and forecast, and limits placed where a reader will see them (1 if all three are at Good or better; 0.5 if all three are present but at least one falls below Good; 0 if any is missing).

Honest accounting is required. The strongest projects are not the ones that prove their model right. They are the ones that say clearly what the data establishes, what it cannot, and what it would take to know more.

Module exercises

Each module has one short exercise, separate from your project. Everyone works on the same shared dataset, and it stays the same across all three modules: you will classify it, then analyze it, then reason causally about it. We build common skills on common ground, and each exercise builds on the last. The exercise has two parts:

1. **Review and implement.** Apply the module's main concepts to the shared data with your AI analyst: classify and describe variables (Module 1), visualize and find associations (Module 2), draw a causal diagram, name the confounders, and forecast the effect of a decision with an honest range (Module 3).
2. **Reflect.** In a page or so, say what the results mean, what they cannot tell you, and where your AI analyst needed correcting.

Exercises are individual and straightforward by design. They check the core ideas so that your project can go further. The Module 1 exercise also includes the **class survey**, a short Google Form due by 6:00 pm Thursday, October 15: it is worth 1 of the exercise's 10 points, credited for any on-time submission.

Exercise	Covers	Due
Module 1 exercise	Kinds of variables, operationalization, data quality, describing a variable	Sun Nov 1 (class survey: Thu Oct 15, 6:00 pm)
Module 2 exercise	Visualization, grouping, association, the limits of correlation, and explaining vs. predicting	Sun Nov 15
Module 3 exercise	Pearl's ladder, causal diagrams, confounders, comparing changes, and a forecast with a range	Sun Nov 22

Assessment

Your grade comes from three module exercises, the project built in stages, and participation. There is no exam.

Component	% of grade
Module exercises (3, 10% each)	30
Project Brief	5
Data & Methods draft (Module 1)	10
Results draft (Module 2)	10
Pre-final version: white paper and slide deck (Dec 3)	5
Final project: video presentation, slide deck, white paper	30
Participation: in-person work, Thursday pod posts, the Roundtable reflection, pod feedback (including your review of pod-mates' pre-final versions), dry run	10

Drafts are graded on the thinking, not the polish: is the model clear, are the variables well chosen, is the account of the evidence honest? Rubrics for every project stage, the pre-final version, the final project, and each module exercise are on Canvas under Files, in the Rubrics folder.

The final project is scored holistically, as one piece of work, from a baseline of 10 points for each of its three parts: the video, the slide deck, and the white paper.

Grade	Points
A	92.5–100
A−	89.5–92.4
B+	87.5–89.4
B	82.5–87.4
B−	79.5–82.4
C	69.5–79.4
F	below 69.5

Course schedule

The Salem cohort meets in person on Tuesdays in Kaneko 120. Thursday online sessions are shared with the Portland cohort (622A-P). Detailed session plans for each module are posted on the course site the week before it begins.

Week of	Session	Date	Topics	Due
Oct 12	Intro (async)	Thu Oct 15	Read Getting Started (on Canvas); watch the intro videos; check your computer setup with TechReady if needed	**Class survey** (6:00 pm); begin your Project Brief
Oct 19	**Module 1** kick-off	**Tue Oct 20**	Data is the measurement of models; kinds of variables; where data lives; operationalization	**Project Brief**
	Online 1	Thu Oct 22	AI as analyst: interrogating a dataset; measures for your own concepts	Pod post
Oct 26	In-person 2	**Tue Oct 27**	Five ways data lies; simulation; describing a variable	Variable table and data-source decision
	Online 2 (recorded)	Thu Oct 29	Describe your own data with AI; Data & Methods draft workshop; Atkinson Community Roundtable with Oregon's Chief AI Officer (6:00 pm)	Pod post; Roundtable reflection; **Module 1 exercise** (Sun Nov 1)
Nov 2	**Module 2** kick-off	**Tue Nov 3**	Visualization: what question each chart answers; grouping; association	**Data & Methods draft**
	Online 1	Thu Nov 5	Hands-on analysis with AI; pod work	Pod post

Week of	Session	Date	Topics	Due
Nov 9	In-person 2	**Tue Nov 10**	Deeper analysis; pods share preliminary findings; back to the model; explaining vs. predicting	
	Online 2	Thu Nov 12	Work session; the limits of correlation	Pod post; **Module 2 exercise** (Sun Nov 15)
Nov 16	**Module 3** kick-off	**Tue Nov 17**	Why correlation is not enough; Pearl's ladder; causal diagrams; confounders	**Results draft**
	Online 1	Thu Nov 19	Causal diagram workshop; what would move you up the ladder; forecasting the effect of a decision	Pod post; **Module 3 exercise** (Sun Nov 22)
Nov 23	Thanksgiving	No class	Reflection week. I am available online. No deadlines.	
Nov 30	Final in-person	**Tue Dec 1**	Project polish; presentation dry run with your pod	**Dry run**
	Final online: pod review	Thu Dec 3	Pod review of pre-final projects; final questions; concluding remarks; course evaluations (6:00–6:20 pm)	**Pre-final version** (6:00 pm)
	Final project due	Fri Dec 4	Video, slide deck, and white paper	**Final project** (11:59 pm)
	LEAD Connect	Sat Dec 5, Salem	Program event, separate from this course's project	

Policies

AI in this course. AI use is expected, not just permitted. It is how we do the technical work. Three rules apply:

1. **Disclose it.** Each submission includes a short note on how you used AI: which tool, for what, and what you changed or rejected.
2. **Check it.** AI is often confident and wrong. It will call an ordered rating a measurement, invent a source, or miss what is odd in a dataset. You are accountable for every number and claim you submit.
3. **Own the thinking.** The model, the choice of what to measure, and the judgment about what the evidence means must be yours. AI does the execution; you do the reasoning.

If you simulate data, say so plainly and state every assumption. Presenting simulated data as real, or tuning a simulation toward the answer you want, is a breach of academic honesty.

Our AI tool. The course standard is **Gemini, signed in with your Willamette account**. Under the university's Google Workspace agreement, what you share there is not used to train models. All demonstrations and prompt cards use Gemini. You may use other tools (Claude, ChatGPT, others) if you disclose it, but personal accounts have different data terms. Check their settings and treat them as public.

Workplace data. A Willamette account protects you, not your employer. Use workplace data only if your employer's policy allows it, remove names and identifying details first, and when in doubt, simulate.

Honesty and integrity. You are expected to abide by the Atkinson Graduate School of Management Honor Code and its application to academic honesty, as described in the current student handbook. Plagiarism is copying material from the Internet, library resources, AI, or other sources without attribution. Drawing on the work of others is fine when you cite it: give references for ideas that are not your own, and use quotation marks for anything taken verbatim. All work you submit must be essentially your own.

Accessible Education Services. Students with disabilities who require accommodation should let me know the nature of the accommodation in the first week of class. Additional support is available from Willamette University Accessible Education Services.

Attendance and participation. The in-person nights carry the course; please attend and come prepared. Your contributions, discussion, and active listening are part of your participation grade. Illness and work required by your employer are acceptable reasons to miss. If you need to join an in-person night remotely, email me in advance; the Google Meet link is listed above. Thursday sessions are recorded, and their activities can be completed on your own time by Sunday night.

Late work. The course moves quickly and each stage builds on the last, so late work is generally not accepted. If you see a conflict coming, ask me before the deadline. Professional managers arrange extensions in advance; they do not negotiate them after the fact.

Questions. Bring them early and often. Email, my open calendar, or Google Meet all work. I eagerly await your puzzles.